

Date _____

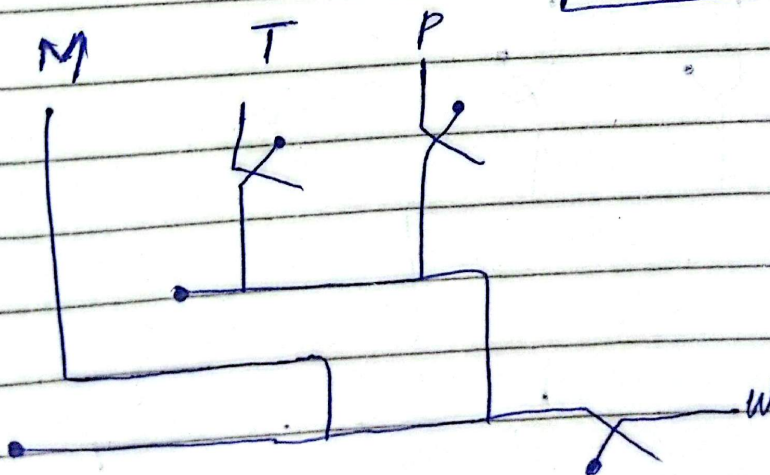
(Q1)

Truth table:

T	P	M	W	
0	0	0	0	$T'P'M'W'$
0	0	1	0	$T'P'MW'$
0	1	0	0	$T'PM'W'$
0	1	1	1	$T'PMW$
1	0	0	0	$TP'M'W'$
1	1	0	1	$TPM'W$
1	0	1	0	$TP'MW'$
1	1	1	1	$TPMW$

$$W = T'P'M + T'PM + TPM' + TP'M + TPM$$

$$W = M + (TP)$$

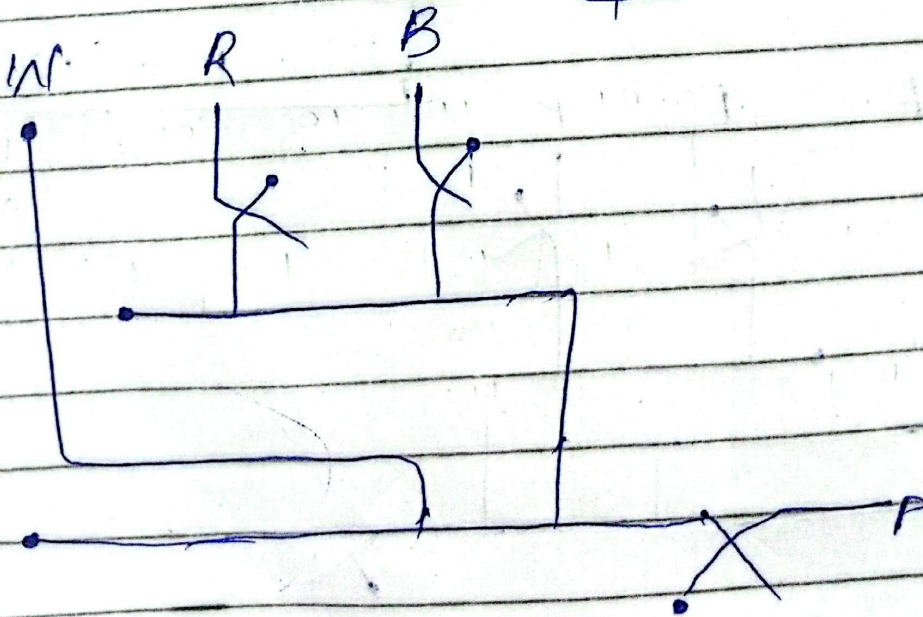


Q2) Truth table:

W	R	B	P	W'	R'	B'	P'
0	0	0	0	1	1	1	1
0	0	1	0	1	1	0	1
0	1	0	0	1	0	1	1
0	1	1	1	1	0	0	0
1	0	0	1	0	1	1	0
1	0	1	1	0	1	0	0
1	1	0	1	0	0	1	0
1	1	1	1	0	0	0	1

$$P = W'R'B + WR'B' + WR'B + WRB' + WRB$$

$$P = W + (R \cdot B)$$



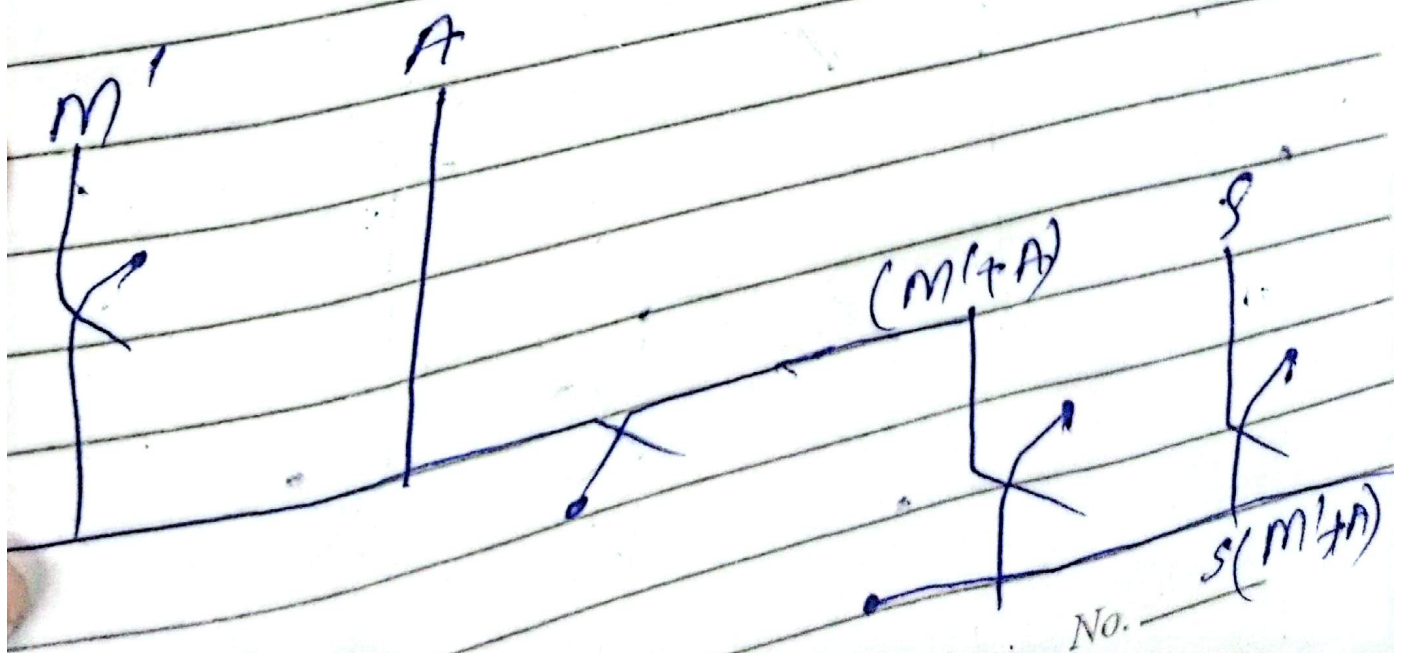
Q3) Truth table:

S	A	M	F	
0	0	0	0	$S'A'M'F'$
0	0	1	0	$S'A'MF'$
0	1	0	0	$S'AM'F'$
0	1	1	0	$S'AMF'$
1	0	0	1	$SA'M'F$
1	0	1	0	$SA'MF'$
1	1	0	1	$SAM'F$
1	1	1	1	$SAMF$

Boolean expression:

$$F = SA'M' + SA M' + SAM$$

$$F = S(M' + A)$$



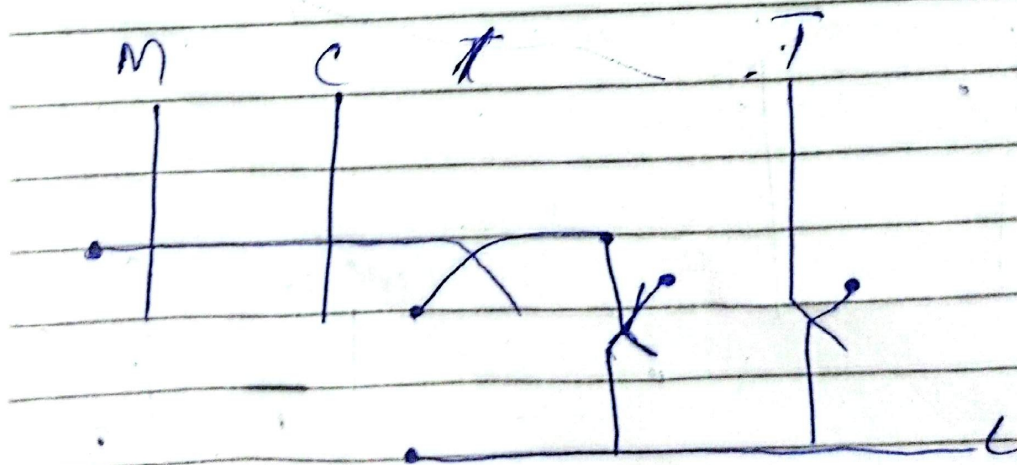
(Q4) Truth table:

M	C	T	L	M' C' T' C'
0	0	0	0	M' C' T' C'
0	0	1	0	M' C' T' C'
0	1	0	0	M' C' T' C'
0	1	1	1	M' C' T' C'
1	0	0	0	M C' T' C'
1	0	1	1	M C' T' C'
1	1	0	0	M C T' C'
1	1	1	1	M C T' C'

Boolean expressions:

$$L = M'CT + MC'T + MCT$$

$$L = T(M+C)$$



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DS

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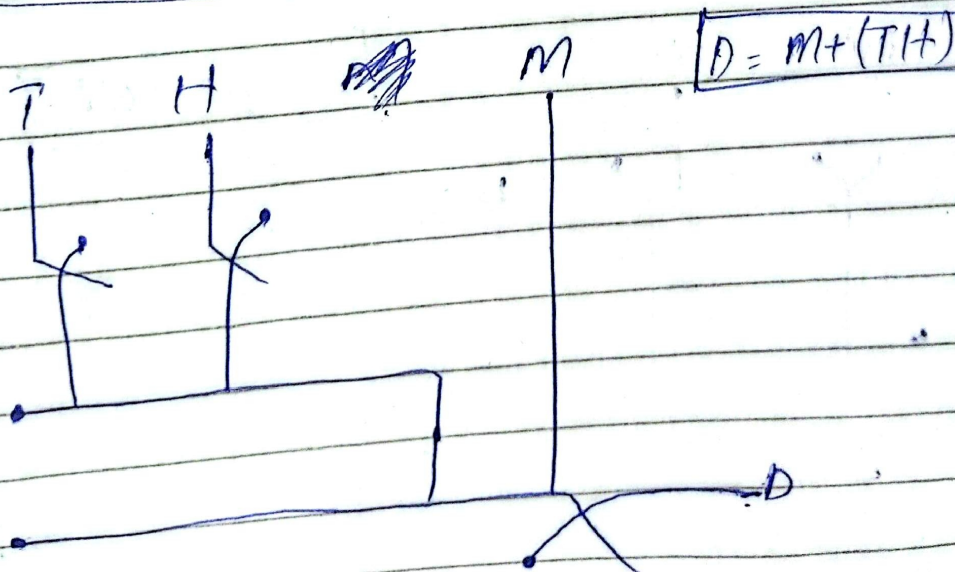
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Q5) Truth Table:

T	H	M	D	
0	0	0	0	$T'H'M'D'$
0	0	1	1	$T'H'MD$
0	1	0	0	$T'H M'D'$
0	1	1	1	$T' H M D$
1	0	0	0	$T H'M'D'$
1	0	1	1	$T H' M D$
1	1	0	1	$T H M' D$
1	1	1	1	$T H M D$

Boolean expression:

$$D = T'H'M + T'HM + TH'M + THM + THM$$



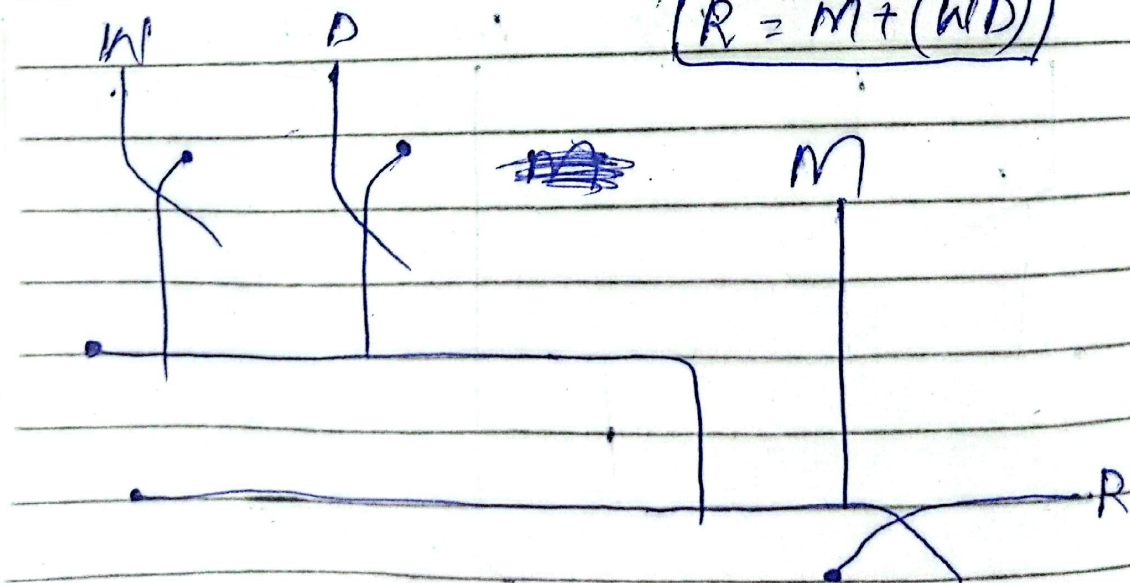
(Q4) Truth table:

W	D	M	R	$W'D'M'R'$
0	0	0	0	$W'D'M'R$
0	0	1	1	$W'D'M'R'$
0	1	0	0	$W'DMR$
0	1	1	1	$W'DM'R'$
1	0	0	0	$W'D'MR$
1	0	1	1	$WDM'R$
1	1	0	1	$WDMR'$
1	1	1	1	$WDMR$

Boolean expression:-

$$R = W'D'M + W'DM + WD'M + WDM' + WDM$$

$$R = M + (WD)$$



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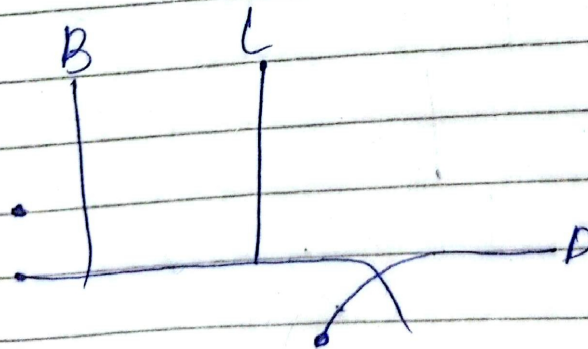
(Q7) Truth table:

L	W	B	D	L'W'B'D'
0	0	0	0	L'W'B'D'
0	0	1	1	L'W'B'D
0	1	0	0	L'WB'D'
0	1	1	1	L'WB'D
1	0	0	1	LW'B'D
1	0	1	1	LW'B'D
1	1	0	1	LWB'D
1	1	1	1	LWB'D

Boolean expressions:-

$$D = L'W'B + L'WB + LW'B' + LW'B + LW'B' + LWB$$

$$D = B + L$$



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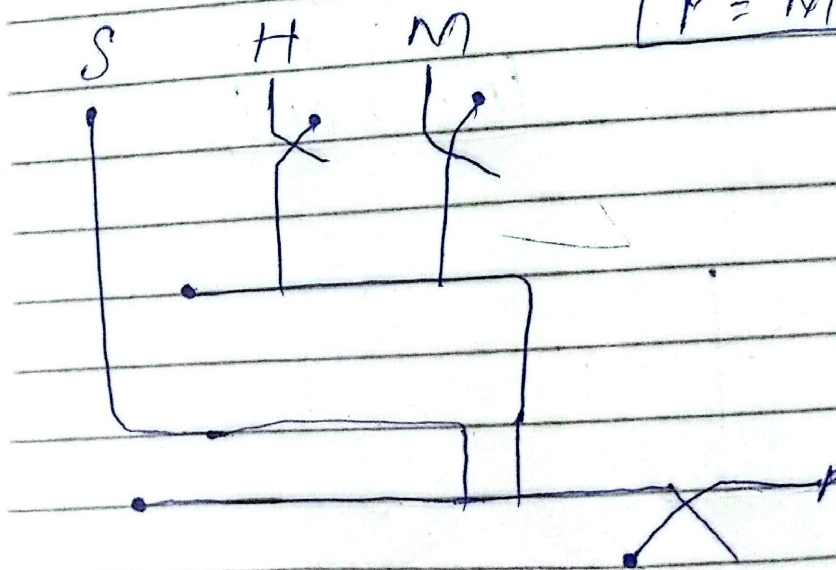
(Q6) Truth Table:

S	H	M	P	
0	0	0	0	$S'H'M'P'$
0	0	1	1	$S'H'M'P$
0	1	0	0	$S'H'M'P'$
0	1	1	1	$S'H'M'P$
1	0	0	0	$S'H'M'P'$
1	0	1	1	$S'H'M'P$
1	1	0	1	$S'H'M'P$
1	1	1	1	$S'H'M'P$

Boolean Expression:

$$P = S'H'M + S'H'M + S'H'M + S'H'M + S'H'M$$

$$P = M + S'H$$



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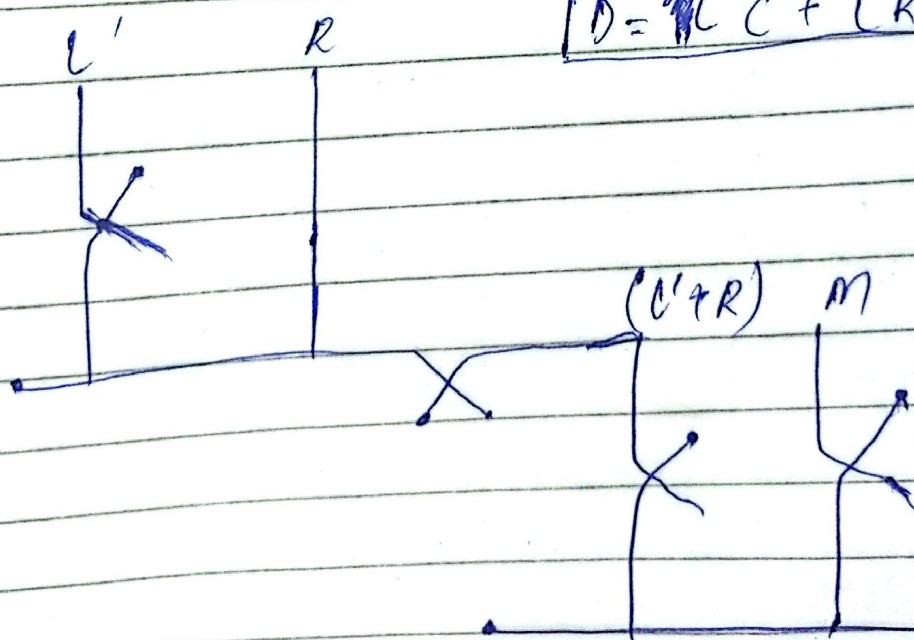
(Q9) Truth table:

L	C	R	D	L' C' R' D'
0	0	0	0	L' C' R' D'
0	0	1	0	L' C' R D'
0	1	0	1	L' C R' D
0	1	1	1	L' C R D
1	0	0	0	L C' R' D'
1	0	1	0	L C' R D'
1	1	0	0	L C R' D'
1	1	1	1	L C R D

Boolean expression =

$$D = L' C R' + L' C R + L C R$$

$$D = L' C + L R C$$



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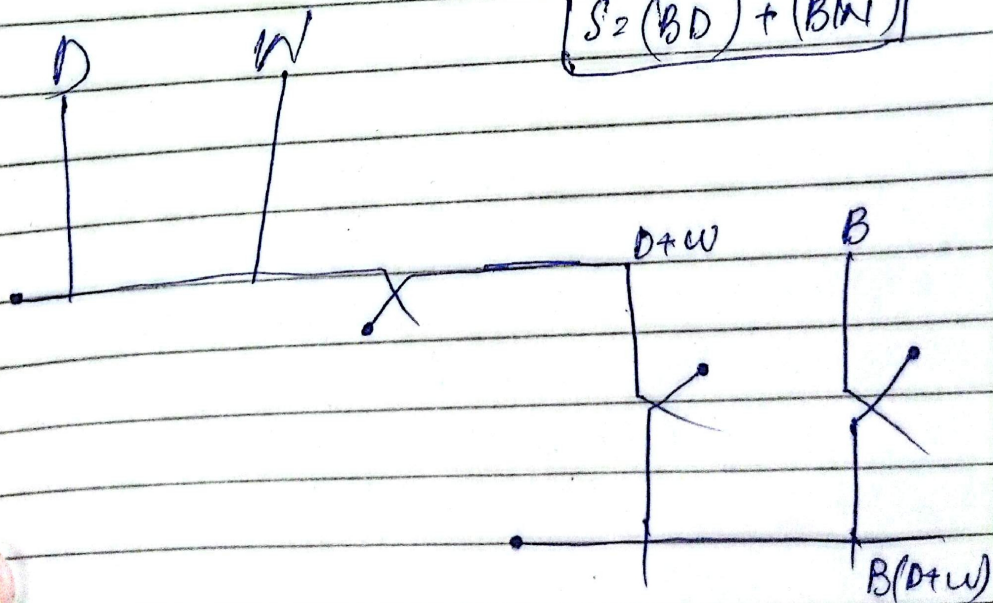
Q10) Truth Table:

B	D	W	S	
0	0	0	0	$B' D' W' S'$
0	0	1	0	$B' D' W S'$
0	1	0	0	$B' D W' S'$
0	1	1	0	$B' D W S'$
1	0	0	0	$B^* D' W' S'$
1	0	1	1	$B D' W S$
1	1	0	1	$B D W' S$
1	1	1	1	$B D W S$

Boolean expression:-

$$J = BD'W + BDW' + BDW$$

$$S_2(BD) + (BM)$$



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